

Beyond the AI Hype
Evaluating LLMs vs. Digits AGL[®] for Accounting Tasks

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1 Executive Summary

1.1 Key Findings

- **Performance Ceiling:** No general-purpose LLM achieved greater than 81% accuracy in one-shot classification, and no model exceeded 87% accuracy when deployed within the agent harness.
- **Human Baseline Surpassed:** 6 general-purpose models have surpassed human accuracy with zero hallucinations, including Claude Fable 5, Claude Opus 4.8, Claude Sonnet 4.6, GPT 5.5, GPT 5.4 Mini and Gemini 3.5 Flash. The top-performing model exceeds human accuracy by 1.6%.
- **Frontier Models Demonstrate Effective Use of Contextual Information:** Compared with one-shot LLM inference, agent-harness runs improved accuracy by 4.5 to 9.1 percentage points, indicating that frontier LLMs can ground decisions in retrieved business context rather than relying only on internal training knowledge. These gains, however, carried substantial operational cost relative to the one-shot LLM baseline, with latency increasing 5x to 22x and token usage increasing 8x to 29x.
- **Specialized System Advantage:** Digits’ purpose-built classification system significantly outperformed all evaluated frontier LLMs across both task types. Digits AGL[®] exceeded the top-performing LLM by 17.1 percentage points in one-shot categorization and by 11 percentage points in multi-turn agentic tasks, while operating at 1,634× lower latency and 311× lower token cost.

1.2 Methodology Overview

- We evaluated 14 state-of-the-art frontier reasoning models, including 5 open-weight and 9 closed-weight models, to categorize transactions using a dataset of 2,000 financial transactions collected from more than 4 randomly selected small businesses.
- Using the human baseline established from the same dataset and 12 outsourced professional accountants, we assessed the accuracy, latency, and hallucination rate of these 14 reasoning models at various thinking effort levels.
- In addition to benchmarking model performance in one-shot categorization tasks, we also tested the top-ranking models in an agent harness to enable more informed classification outcomes based on historical transactions and internet search results.

2 Introduction

In the financial technology ecosystem, the application of artificial intelligence to accounting automation has garnered significant attention, accompanied by both substantial technical challenges and industrial hype. As organizations increasingly prioritize automating accounting tasks, a rigorous assessment of the capabilities and inherent limitations of large language models (LLMs) becomes imperative. This study presents a multi-dimensional evaluation of Digits’ specialized machine learning architecture, i.e. Digits Agentic General Ledger[®] (Digits AGL[®]), against 14 frontier general-purpose LLMs, which are assessed both as standalone solutions and in conjunction with agent harness architectures. The evaluation focuses on accuracy, latency, reliability, and token usage metrics.

While recent iterations of LLMs and agentic setups have demonstrated remarkable proficiency across diverse domains, their efficacy in addressing the highly nuanced and subjective requirements of professional accounting classification remains a critical area of inquiry.

3 Background

Accounting practices are inherently subjective, with significant variations in how individual accountants approach classification and decision-making processes. This subjectivity presents a fundamental challenge for LLMs attempting to generalize accounting knowledge globally. While these models can learn general accounting principles, they struggle to replicate the nuanced judgment experienced accountants develop through years of practice in specific industry contexts. This limitation becomes particularly evident when models trained on generalized data attempt to mimic the decision-making patterns of individual accountants working within specialized domains.

The accounting industry is also rapidly embracing AI agents as a means of embedding domain knowledge and firm-specific practices directly into automated workflows. These agentic systems hold the promise of going beyond static model inference by dynamically incorporating an individual firm’s internal guidelines, historical decisions, and contextual reasoning into the classification process. This adoption introduces critical questions regarding the efficacy of such systems, particularly how well they can navigate the subjective nature of accounting, as well as the overall cost-effectiveness compared to traditional methods and standalone LLMs.

In light of these challenges, we were interested in how Digits’ proprietary machine-learning system, designed specifically for financial transaction classification, compares against state-of-the-art LLMs. In particular, this study seeks to understand the costs and benefits of using frontier LLMs in both one-shot category classification tasks and in conjunction with an agent harness.

4 Methodology

Transaction categorization serves as a cornerstone of accounting, providing the foundation for precise financial records and critical business insights. Consequently, our evaluation primarily focused on predicting the correct category from a business’s unique Chart of Accounts (CoA) for any given

transaction.

Predicting categories of a business-specific CoA better reflects real-world accounting practices where businesses maintain individualized category structures. To provide context for the classification task, we supplied the models with the “other side” of each transaction, as this information would typically be available from their bank feed, the credit card provider, payroll partner, or other financial integration source.

4.1 Data and Ground Truth

Our study utilized a dataset comprising 2,000 financial transactions collected from 4 businesses that use Digits, which was the same dataset used to establish a human baseline. The dataset reflected typical transaction patterns with an 88% to 12% split between debits and credits, mirroring the natural distribution commonly observed in accounting systems. The split is skewed towards debit transactions because most transactions’ credit side can be defined by their source (e.g., a given bank account). The number of categories in the Charts of Accounts varied significantly across these businesses, ranging from 15 to 128.

US-based GAAP accountants have reviewed all transactions and expected categories, providing a solid basis for this comprehensive comparison.

4.2 Establishing a Human Baseline

In June 2025, our study involved 12 outsourced accounting participants, ranging from seasoned accountants with decades of experience to college seniors in Accounting. We organized these individuals into four distinct teams of three. Each team was assigned 500 transactions from a single business within our benchmark dataset for independent categorization. To determine the final classification for each sample, we utilized a majority-vote system within each three-person group.

Critically, all outsourced accountants received identical information sets to those provided to the LLMs being evaluated, ensuring consistency in available data across both human and artificial intelligence classification approaches.

4.3 Model Providers

We evaluated 14 models with various reasoning configurations from all major providers, including OpenAI, Anthropic, Google, DeepSeek, Moonshot AI, and Z.ai. The complete model list is shown in Table 1. To standardize deployment for open-source models, we leveraged Baseten’s infrastructure and exclusively used the OpenAI client SDK for all model connections, avoiding provider-specific SDKs. The reasoning models were evaluated at various effort levels to determine their impact on performance metrics.

Provider	Model configurations
Anthropic	Claude Fable 5 (low, medium, high, xhigh, max)
	Claude Opus 4.8 (thinking disabled, low, medium, high, max)
	Claude Opus 4.7 (thinking disabled, low, medium, high)
	Claude Sonnet 4.6 (thinking disabled, medium)
OpenAI	GPT-5.5 (thinking disabled, low, medium, high, xhigh)
	GPT-5.4 mini (low, medium, high)
	GPT OSS 120B (thinking disabled, medium)
Google	Gemini 3.1 Pro Preview (low, medium, high)
	Gemini 3.1 Flash-Lite (minimal, low, medium, high)
	Gemini 3.5 Flash (minimal, low, medium, high)
	Gemma 4 31B (minimal, medium)
DeepSeek	DeepSeek-V4 Pro (high, max)
Moonshot AI	Kimi-K2.6 (thinking disabled, medium)
Z.ai	GLM-5 (thinking disabled, medium)

Table 1: List of evaluated models by provider

4.4 One-Shot Categorization

During this phase of the study, we assessed the performance of Large Language Models (LLMs) in a single-turn interaction to perform transaction classification. Each model was provided with a uniform system and user prompt, directing it to produce a response in a structured JSON format. To ensure consistency across different inference providers, we standardized the construction of all LLM requests by utilizing the OpenAI client SDK.

All models received the standardized system prompt: "You are an expert bookkeeper with deep knowledge of accrual-accounting and an eye for detail."

The user prompt template was dynamically populated with transaction-specific data, including the transaction description, monetary amount, the known counter-entry originating from the ingestion source, and the business’s complete Chart of Accounts excluding the category associated with the known counter-entry.

4.4.1 Prompt Setup

Prompt 1 shows an example of the prompt we used for the model evaluation.

4.4.2 Prompt Parameters

All evaluated LLMs in this revision rejects `temperature`, `top_p`, or `budget_tokens`. Consequently, these inference parameters were excluded from the standardized prompt configuration.

Prompt 1: Example Prompt Template for Model Evaluation

- 1: Description: UBER *TRIP
 - 2: Merchant name: Uber
 - 3: Amount: \$44.80
 - 4: Given that this will be recorded as a credit to Mercury Credit (0000) - 1
 - 5: Which category should receive the debit side?
 - 6: "Software & Apps"
 - 7: "Travel"
 - 8: "Meals"
 - 9: <other categories removed from publication for privacy reasons>
 - 10: * I want you to think of the most likely category from the list above for the described transaction and amount
 - 11: * Think of a single sentence description of why
 - 12: * Double check that the category is actually in the list
 - 13: NOTE: Do not provide explanations. Only provide the most relevant category.
 - 14: Return them as JSON with the schema: {"category": <string>}
-

4.5 LLMs with Agent Harness

The emergence of general-purpose workplace agents such as Claude Cowork and ChatGPT Codex has demonstrated the considerable potential of reasoning models operating in conjunction with well-designed tool ecosystems. These systems have shown that augmenting model inference with structured access to external resources and multi-step reasoning can yield substantially stronger performance than static, single-pass approaches.

In this study, we introduced an agent harness to evaluate the top-ranking LLMs from the one-shot categorization tasks. The harness was intentionally kept lightweight in scope, with the primary objective of equipping reasoning models with targeted retrieval tools capable of surfacing contextual information relevant to transaction booking decisions.

The harness has the following characteristics:

- It has 2 function calls defined:

```
- google_search(query, max_results=5)
- get_historical_transactions(
    category_name=None,
    legal_entity_id=None,
    transaction_timestamp=None,
    entry_type=None,
    max_results=100,
    category=None,
)
```
- Agent runs are capped at four turns where each turn could yield parallel function calls or simply invoking the LLM, after which the model must produce a final classification.

To ensure comparability, all models were evaluated via the OpenAI SDK Chat Completions API. An exception was made for GPT-5 series models, which were evaluated via the Responses API due to the Chat Completions API’s lack of support for function tools with `reasoning_effort`.

4.6 Performance Analysis

This study evaluates model performance across three primary metrics: accuracy, latency, and hallucination rate, each averaged at the task level. For models evaluated within the agent harness condition, average token consumption per task is additionally reported as a fourth metric.

Accuracy serves as the most direct indicator of model utility, reflecting the proportion of transactions correctly classified against ground-truth labels. Latency is equally consequential in applied settings, as accounting practices increasingly demand real-time financial intelligence to support timely decision-making. Hallucination rate is considered as a measure of output trustworthiness, a critical property in high-stakes financial contexts where erroneous or fabricated classifications can have material downstream consequences. Finally, token consumption is examined as a proxy for operational cost, given that the economic viability of deploying LLM-based solutions at scale is a primary consideration for firms seeking automation solutions.

In this study, a hallucination is defined as any time a model suggests a category that is not in the business’s actual Chart of Accounts. We treated any such outside category as a clear error (a “true hallucination”) rather than assuming the Chart of Accounts was simply missing a category.

4.6.1 One-Shot Categorization Performance

6 general-purpose LLMs in this evaluation outperformed outsourced human accountants in accuracy in the one-shot category classification task. In previous revisions of this study, no LLMs exceeded the human accuracy baseline. The models that have surpassed the human accuracy baseline are: Claude Fable 5, Claude Opus 4.8, Claude Sonnet 4.6, GPT 5.5, GPT 5.4 Mini, and Gemini 3.5 Flash. Specifically, Claude Opus 4.8 (medium reasoning effort) surpassed the human accuracy baseline by 1.6%.

Our comprehensive evaluation of the latest frontier models revealed improved performance across model providers. Top-ranking models generally have higher accuracy compared to our January 2026 study. The average model accuracy in this study was improved from 73.4% to 76.8% compared to our previous benchmark result on this dataset. This 3.4% improvement indicates overall progress in general-purpose LLMs across major providers.

Average latency across evaluated models in the one-shot categorization task declined from 6.25 to 3.26 seconds per task, reflecting continued inference efficiency gains across the frontier model landscape. Notably, Claude Opus 4.8 achieved response times below 1.61 seconds across all reasoning effort levels, representing the lowest latency observed among the models evaluated.

Hallucination rates across evaluated models declined notably relative to the prior benchmark revision, falling from 0.07% to 0.03%, suggesting broad improvements in instruction-following alignment across model providers.

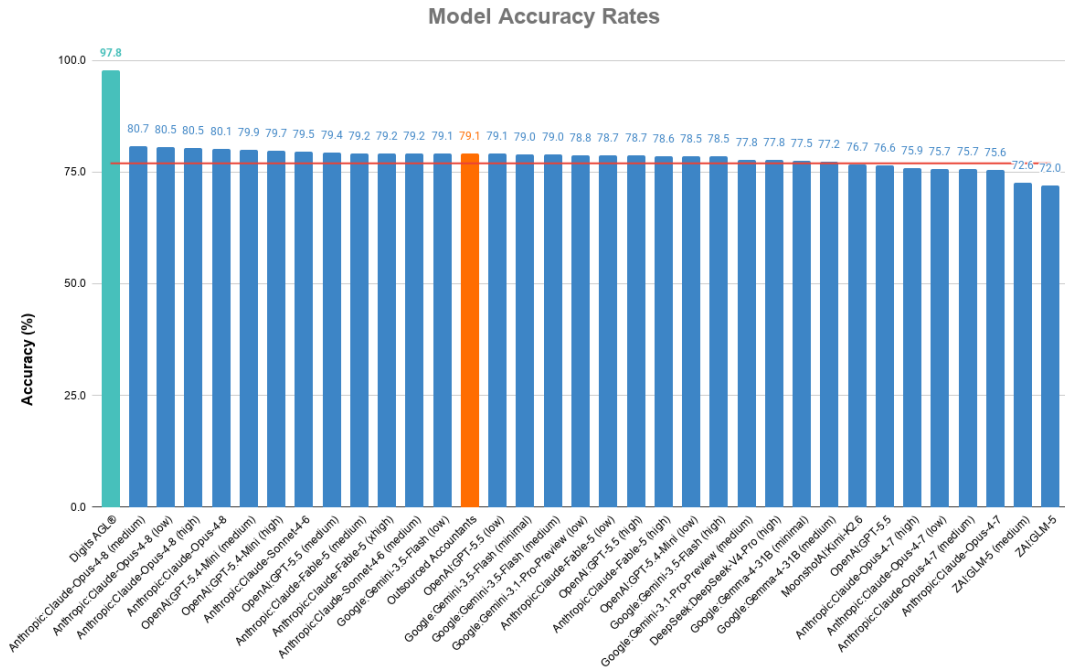


Figure 1: Comparison of model accuracy rates in the one-shot categorization task (higher is better)

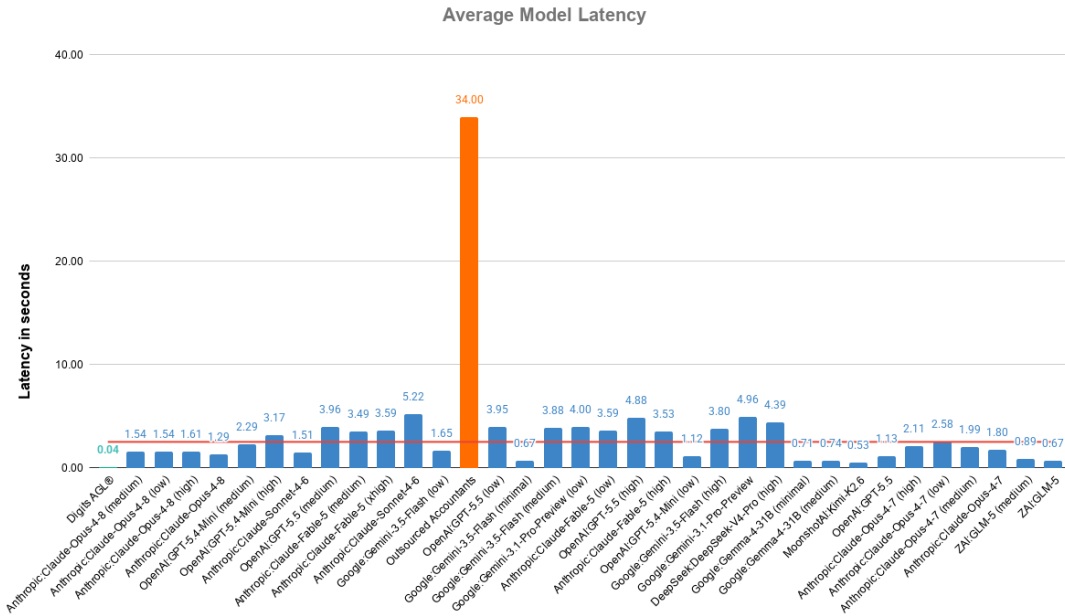
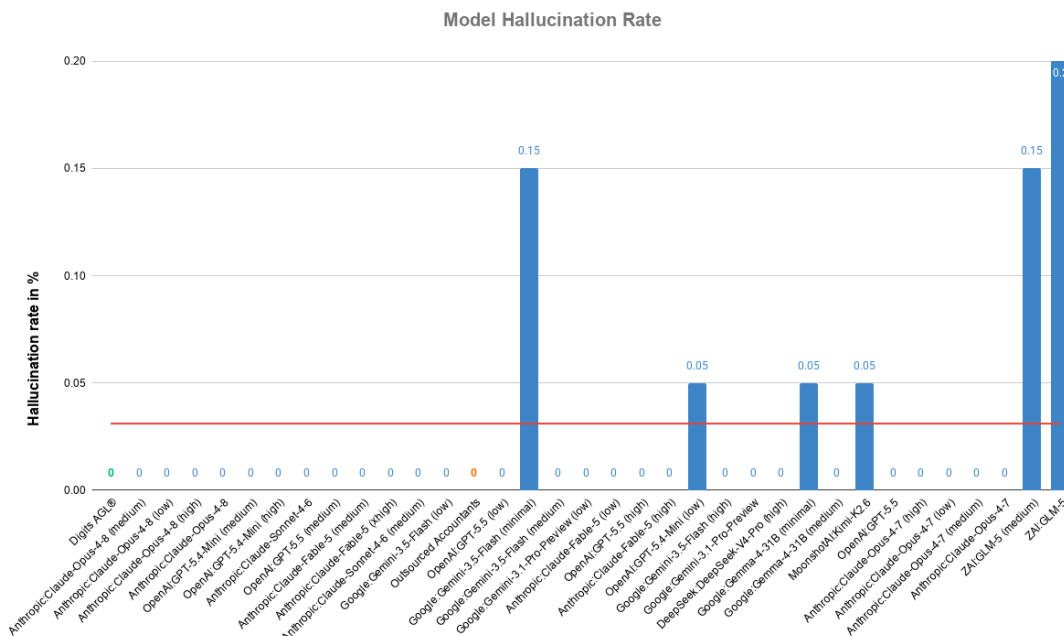


Figure 2: Average model latency in the one-shot categorization task (smaller is better)



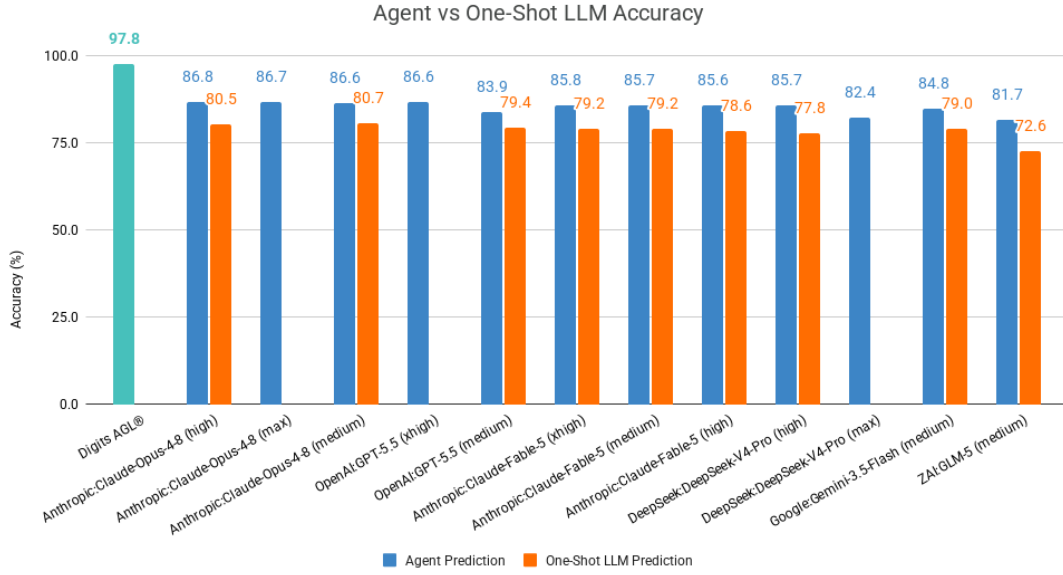


Figure 4: Accuracy comparison between one-shot categorization and agent harness conditions (higher is better)

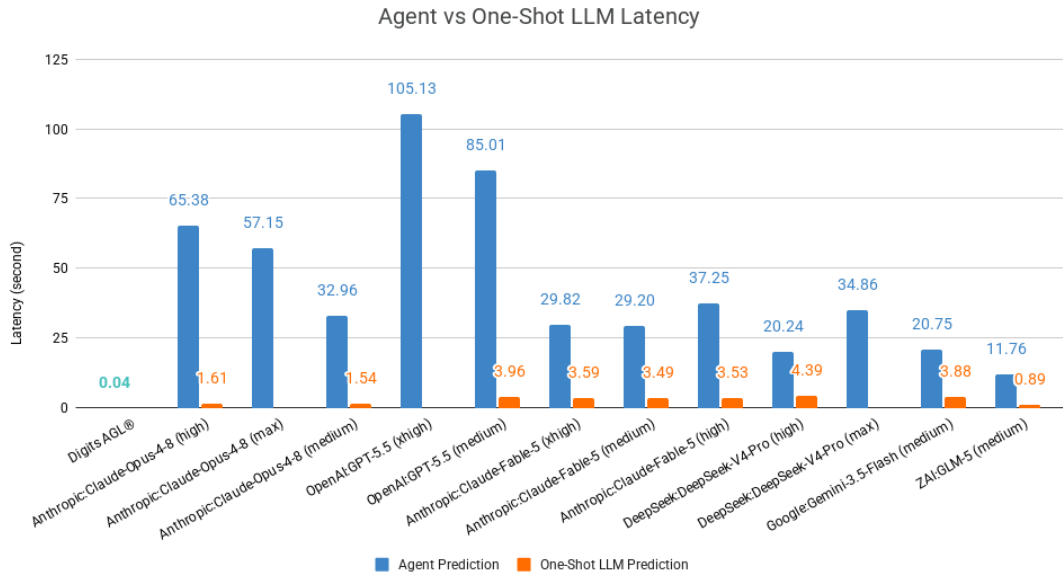


Figure 5: Latency comparison between one-shot categorization and agent harness conditions (smaller is better)

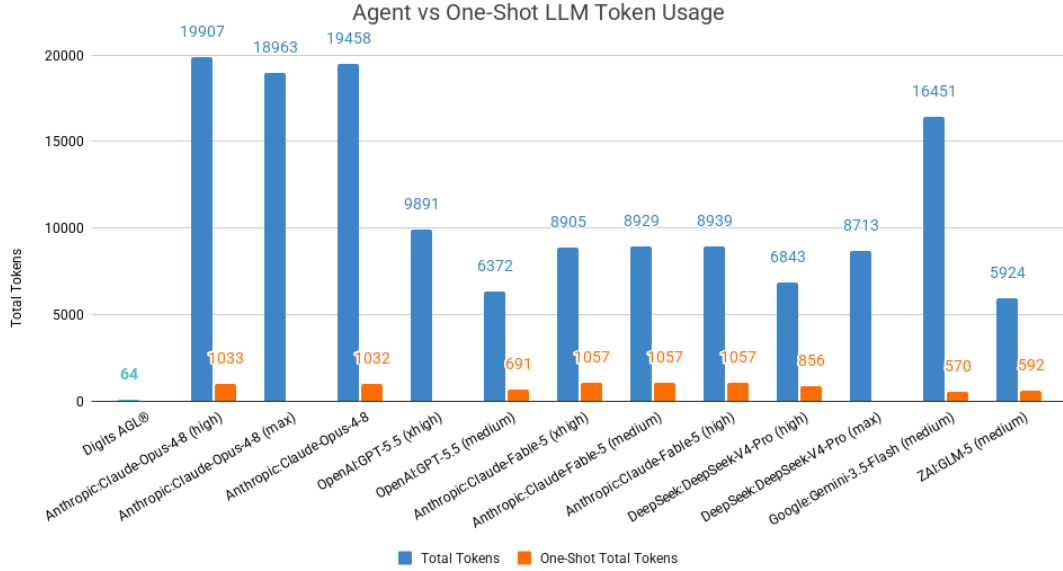


Figure 6: Token usage comparison between one-shot categorization and agent harness conditions (smaller is better)

4.6.3 Latency Comparison: One-Shot Categorization vs. Agent Harness

Figure 5 illustrates the latency cost associated with the accuracy gains observed under the agent harness condition relative to one-shot inference, with per-task response times increasing by 5 to 22 fold across evaluated reasoning models.

Of particular note is the performance of DeepSeek V4 Pro within the agent harness condition, which yielded an accuracy improvement of 7.9 percentage points over its one-shot baseline. This gain positioned DeepSeek V4 Pro as the second most accurate LLM evaluated under the agent harness condition, surpassing closed-weight models including GPT-5.5 and Gemini 3.5 Flash, while operating at approximately two-thirds of the inference latency of the top-performing model. These results suggest that open-source reasoning models, when provided with structured contextual scaffolding, are increasingly competitive with their closed-weight counterparts on domain-specific classification tasks.

4.6.4 Token Usage Comparison: One-Shot Categorization vs. Agent Harness

Associated with the inference time increase, the agent harness condition also imposed a substantial increase in token consumption relative to one-shot inference, with per-task token usage rising by 8 to 29 fold across evaluated models. This elevated consumption directly translates to higher operational costs at scale.

Critically, despite trailing Digits AGL® by only 11 percentage points in accuracy, Claude Opus 4.8 at high reasoning effort requires 1,634 times the latency and 311 times the token consumption of Digits AGL®, which completes each task in 40 milliseconds at an average of 64 tokens. Differences

of this magnitude raise significant concerns about the cost-effectiveness of general-purpose LLMs in high-volume production environments.

4.6.5 Hallucination Assessment

All models evaluated under the agent harness condition produced zero hallucinations, compared to a mean hallucination rate of 0.03% observed in the one-shot setting. This finding suggests that providing reasoning models with additional turns and enriched contextual information meaningfully reinforces adherence to the decision scope defined in the task instruction.

5 Comparison with Human Baseline

The human baseline evaluation, conducted across 2,000 transactions spanning four distinct businesses and reviewed by 12 outsourced accountants, yielded several significant insights into the consistency and efficiency of human classification at scale. Against the GAAP accountant-reviewed ground-truth dataset, the outsourced accountants achieved an aggregate accuracy of 79.1%. Notably, 10.4% of transactions exhibited classification disagreement within the three-person accountant groups, reflecting the inherent ambiguity and interpretive complexity present in real-world transaction classification tasks. With respect to efficiency, accountants required an average of 4 hours and 43 minutes to complete their allocation of 500 transactions, corresponding to approximately 34 seconds per transaction.

6 Cost and Benefits of Reasoning Models

Using the top-ranking models and their reasoning effort from the one-shot categorization tasks as the starting point, we evaluated how each model performed with an agent harness under the same or increased reasoning effort.

Model	Effort	Accuracy (%)	Latency (s)	Tool Calls	Input Tokens	Output Tokens	Total Tokens
Anthropic:Claude-Opus-4-8	medium	86.6	32.96	2.66	19,105	353	19,458
	high	86.8	65.38	2.69	19,552	355	19,907
	max	86.7	57.15	2.66	18,601	363	18,963
Anthropic:Claude-Fable-5	medium	85.7	29.20	2.15	8,521	408	8,929
	high	85.6	37.25	2.15	8,531	408	8,939
	xhigh	85.8	29.82	2.15	8,500	405	8,905
OpenAI:GPT-5.5	medium	83.9	85.01	3.96	6,132	241	6,372
	xhigh	86.6	105.13	4.90	9,418	474	9,891
DeepSeek:DeepSeek-V4-Pro	high	85.7	20.24	3.26	6,258	585	6,843
	max	82.4	34.86	3.93	8,000	713	8,713

Table 2: Reasoning-effort comparison for top-performing models under the agent harness

The reasoning-effort comparison presents accuracy, latency, tool usage, and token usage for the same models evaluated across varying reasoning effort levels. For Claude Opus 4.8, increasing reasoning effort beyond medium yielded diminishing returns, with high effort approximately doubling latency for marginal accuracy gains, and maximum effort producing lower accuracy than either prior setting. Claude Fable 5 exhibited a similar non-monotonic pattern: increasing reasoning effort did not consistently improve accuracy, as high effort slightly reduced performance relative to medium effort, while extra-high effort produced only a marginal gain. In the case of DeepSeek V4 Pro, maximum reasoning effort failed to produce any measurable accuracy improvement, yet incurred meaningfully higher latency and token consumption. For GPT-5.5, enabling extra high reasoning effort yielded a 2.7 percentage point improvement in accuracy, accompanied by a modest 24% increase in latency and a 55% increase in token usage.

These results indicate that the relationship between reasoning effort and classification accuracy is neither consistent nor proportional across model families. The substantial latency and token overhead associated with elevated reasoning effort levels does not reliably translate into performance gains, suggesting that default or intermediate reasoning configurations may represent the more cost-effective operating point for transaction categorization tasks in practice.

Claude Opus 4.8 at high reasoning effort achieved the highest accuracy among all evaluated models at 86.8%, yet fell short of Digits AGL[®] by 11 percentage points. However, this superior accuracy was attained at substantial cost: Claude Opus 4.8 incurred 1,634 $\times$ greater latency and 311 $\times$ higher token cost than Digits AGL[®].

7 Comparison with Digits AGL[®]

Digits AGL[®] employs a combination of purpose-built machine learning models and frontier LLMs, with the proprietary components trained and hosted entirely in-house to preserve data security and maintain system integrity. Of particular note is the system’s approach to fast inference of 40 milliseconds per task and minimal token usage. Digits’ specialized ML framework is engineered to ensure precision and cost efficiency within critical financial environments.

Across both one-shot transaction categorization and agent harness-augmented classification evaluations in this study, Digits AGL[®] consistently outperformed all frontier general-purpose LLMs. This performance advantage is especially meaningful in the context of production accounting workflows, where accuracy, low latency, and cost efficiency are not discretionary qualities but operational requirements. The results suggest that domain-specific systems, trained on task-aligned data and architected with accounting constraints in mind, retain a substantial edge over generalized models despite the rapid capability gains observed across the broader LLM landscape.

8 Future Work

This paper presents the fourth revision of our industry-recognized study examining general-purpose LLMs for accounting automation. We acknowledge that emerging innovations may effectively address several challenges identified in our current evaluation framework, particularly inference speed, token

cost, domain-specific understanding of accounting principles, and maturing of agent harnesses in the accounting ecosystem.

We are committed to helping accounting professionals make informed decisions as AI technologies mature. We will continue to update our benchmark as new frontier models are released, ensuring that our findings remain relevant and actionable for practitioners in the field.

9 Conclusion

This study examined the applicability of frontier large language models to accounting automation through a comprehensive evaluation of 14 state-of-the-art general-purpose LLMs on transaction categorization tasks, assessed both in isolation and within an agent harness framework. The findings consistently point to the limitations of generalized models in replicating the domain-specific judgment required for accounting tasks.

Performance Ceiling and Contextual Limitations. General-purpose LLMs demonstrated a clear accuracy ceiling of 81% under one-shot conditions and 87% when augmented with an agent harness. These bounds suggest that broad world knowledge, even when supplemented with retrieval-augmented context, is insufficient to capture the business-specific patterns, industry conventions, and firm-level bookkeeping preferences that underpin accurate transaction classification. The persistence of this ceiling across architecturally diverse model families reinforces the conclusion that the limiting factor is not model capability per se, but rather the absence of deeply embedded domain context.

Tool-Grounded Reasoning Is Becoming a Frontier-Model Strength. The newest frontier models evaluated in this study demonstrated meaningful gains when given access to lightweight retrieval tools rather than relying solely on pretrained knowledge. Under the agent harness, every evaluated model improved accuracy by 4.5 to 9.1 percentage points and produced zero hallucinations, indicating that these models can incorporate external business context into their decisions. Top agent-harness configurations also made repeated tool calls, from approximately 2.2 to 5.0 calls per task, suggesting that the models were actively using the harness to gather evidence before classifying transactions.

Operational Viability Concerns. Beyond accuracy, the operational profiles of LLM-based approaches raise substantive concerns about production viability at scale. Mean per-task latency of 3.26 seconds under one-shot inference and 34.14 seconds under agent harness conditions, compounded by API reliability considerations, presents meaningful scalability challenges for organizations processing high transaction volumes. These characteristics are particularly consequential during peak financial periods where processing throughput and response time are operationally critical.

Limited Returns on Reasoning Effort. Contrary to common intuition, increasing model reasoning effort did not produce consistent or proportional accuracy gains on transaction categorization tasks. Across both closed- and open-source architectures, elevated reasoning configurations delivered marginal improvements in classification performance while incurring substantially higher latency and token costs. These results suggest that the accuracy ceiling observed in this study is driven primarily by the absence of domain-specific context rather than by any deficiency in reasoning capacity.

Human Baseline Exceeded. Six frontier LLMs surpassed the 79.1% accuracy baseline established by outsourced human accountants on the transaction classification task, a noteworthy result given that both the models and human annotators were evaluated without access to historical business context. This finding underscores the capacity of general-purpose models to match and exceed human-level performance on general ledger coding under equivalent, context-free conditions.